StudySapiens Board MCQs

Chapter 4 — Biomolecules

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. The most abundant compound in living organisms is:

(a) Water
(b) Carbohydrate
(c) Protein
(d) Lipid

2. Organic molecules make up approximately what percentage of the dry mass of living organisms?

(a) 25%
(b) 70%
(c) 50%
(d) 93%

3. The basic element of organic molecules is:

(a) Nitrogen
(b) Carbon
(c) Oxygen
(d) Phosphorus

4. A polymer is made of many repeating units called:

(a) Ions
(b) Isomers
(c) Monomers
(d) Isotopes

5. Proteins are polymers of:

(a) Nucleotides
(b) Fatty acids
(c) Monosaccharides
(d) Amino acids

6. Nucleic acids are polymers of:

(a) Nucleotides
(b) Amino acids
(c) Glucose
(d) Glycerol

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7. A covalent bond is formed by:

(a) Hydrophobic clustering
(b) Transfer of electrons
(c) Sharing of electrons
(d) Hydrogen attraction

8. An ionic bond is formed by:

(a) Hydrogen bonding
(b) Van der Waals forces only
(c) Electrostatic attraction between oppositely charged ions
(d) Sharing of electrons

9. Hydrogen bonds are important in maintaining the structure of:

(a) Only lipids
(b) Only inorganic salts
(c) Proteins and nucleic acids
(d) Only carbohydrates

10. Hydrophobic interactions are important in:

(a) Protein folding and lipid bilayer formation
(b) Ionic bond formation
(c) DNA replication only
(d) Peptide bond formation

11. Condensation reactions build polymers by:

(a) Adding water
(b) Removing water
(c) Removing carbon
(d) Adding oxygen

12. Hydrolysis breaks polymers by:

(a) Adding water
(b) Removing nitrogen
(c) Removing water
(d) Adding phosphate

13. The formation of maltose from two glucose molecules is an example of:

(a) Oxidation
(b) Reduction only
(c) Condensation
(d) Hydrolysis

14. Water is an excellent solvent because it is:

(a) Nonpolar
(b) Polar and forms hydrogen bonds
(c) Acidic
(d) Basic only

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15. Hydrophilic substances are those that:

(a) Are lipids only
(b) Are gases only
(c) Dissolve readily in water
(d) Repel water

16. Hydrophobic substances such as lipids:

(a) Dissolve readily in water
(b) Are always ionic
(c) Do not dissolve in water
(d) Form hydrogen bonds with water

17. The high specific heat capacity of water helps organisms by:

(a) Storing genetic information
(b) Forming cell walls
(c) Stabilizing temperature
(d) Increasing evaporation only

18. The high heat of vaporization of water helps in:

(a) Cell wall formation
(b) Protein synthesis
(c) Cooling by sweating/transpiration
(d) DNA replication

19. Cohesion of water molecules is due to:

(a) Ionic bonding
(b) Covalent bonding within water
(c) Peptide bonding
(d) Hydrogen bonding

20. The ascent of sap in plants is helped by:

(a) Active transport only
(b) Diffusion only
(c) Osmosis only
(d) Cohesion and transpiration pull

21. Monosaccharides are:

(a) Nucleotides
(b) Polymers of amino acids
(c) Lipids
(d) Simple sugars

22. Glucose and fructose are:

(a) Isomers
(b) Isotopes
(c) Polymers
(d) Ions

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23. Maltose is composed of:

(a) Glucose + fructose
(b) Glucose + galactose
(c) Glucose + glucose
(d) Fructose + galactose

24. Lactose is composed of:

(a) Glucose + galactose
(b) Glucose + fructose
(c) Glucose + glucose
(d) Galactose + fructose

25. Sucrose is composed of:

(a) Glucose + fructose
(b) Fructose + galactose
(c) Glucose + galactose
(d) Glucose + glucose

26. The storage carbohydrate in plants is:

(a) Cellulose
(b) Starch
(c) Glycogen
(d) Chitin

27. The storage carbohydrate in animals is:

(a) Cellulose
(b) Glycogen
(c) Starch
(d) Chitin

28. Cellulose is a structural polysaccharide found in:

(a) Plant cell walls
(b) Fungal cell walls
(c) Animal cell membranes
(d) Bacterial cell walls

29. Chitin is found in:

(a) Fungal cell walls and arthropod exoskeletons
(b) Animal muscles
(c) Plant cell walls only
(d) Viral capsids

30. Amylose is:

(a) A highly branched glucose polymer
(b) A protein
(c) An unbranched glucose polymer
(d) A lipid

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31. Amylopectin is:

(a) A branched glucose polymer
(b) A amino acid chain
(c) An unbranched glucose polymer
(d) A nucleotide chain

32. Glycogen is more branched than:

(a) Starch
(b) Cellulose only
(c) Chitin only
(d) DNA

33. The bond between monosaccharides is called:

(a) Ester bond only
(b) Peptide bond
(c) Glycosidic linkage
(d) Phosphodiester bond

34. The basic unit of proteins is:

(a) Amino acid
(b) Monosaccharide
(c) Nucleotide
(d) Fatty acid

35. An amino acid contains an amino group, a carboxyl group, a hydrogen atom and a variable:

(a) R group
(b) Phosphate group
(c) Sugar group
(d) Haem group

36. The bond between amino acids is called:

(a) Hydrogen bond only
(b) Peptide bond
(c) Glycosidic bond
(d) Phosphodiester bond

37. The primary structure of a protein is its:

(a) Alpha helix
(b) Three-dimensional folding
(c) Multiple subunits
(d) Amino acid sequence

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38. Alpha-helix and beta-pleated sheet are examples of:

(a) Quaternary structure
(b) Primary structure
(c) Secondary structure
(d) Tertiary structure

39. The overall three-dimensional shape of a polypeptide is its:

(a) Tertiary structure
(b) Primary structure
(c) Secondary structure
(d) Quaternary structure only

40. Haemoglobin has quaternary structure because it:

(a) Is a lipid
(b) Contains multiple polypeptide chains
(c) Has only alpha helices
(d) Is a single amino acid

41. Sickle cell haemoglobin results from a change in:

(a) Amino acid sequence
(b) Fatty acid sequence
(c) Nucleotide sequence only
(d) Sugar sequence

42. Fibrous proteins are generally:

(a) Structural and insoluble
(b) Hormones only
(c) Soluble enzymes
(d) Antibodies only

43. Globular proteins are generally:

(a) Only genetic material
(b) Structural and insoluble
(c) Functional and soluble
(d) Only membrane lipids

44. Enzymes are chemically mostly:

(a) Nucleic acids only
(b) Lipids
(c) Carbohydrates
(d) Proteins

45. Lipids are insoluble in water because they are:

(a) Polar
(b) Ionic
(c) Proteins
(d) Nonpolar

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46. Triacylglycerols are composed of:

(a) Glycerol and phosphate
(b) Glycerol and three fatty acids
(c) Nucleotides
(d) Amino acids

47. Saturated fats are usually:

(a) Solid at room temperature
(b) Liquid at room temperature
(c) Gases
(d) Soluble in water

48. Unsaturated fats are usually:

(a) Proteins
(b) Solid at room temperature
(c) Liquid at room temperature
(d) Insoluble in all solvents

49. Phospholipids are important components of:

(a) Ribosomes
(b) Cell membranes
(c) Cell walls only
(d) Chromosomes

50. Steroids have a structure composed of:

(a) Four fused carbon rings
(b) Amino acid chains
(c) Sugar rings only
(d) Long fatty acid chains only

51. Cholesterol is a:

(a) Carbohydrate
(b) Steroid
(c) Nucleic acid
(d) Protein

52. Prostaglandins are involved in:

(a) Genetic storage
(b) Inflammation and local signalling
(c) Cell wall formation
(d) Oxygen transport

53. Aspirin reduces inflammation by inhibiting the synthesis of:

(a) Insulin
(b) Cellulose
(c) Prostaglandins
(d) Haemoglobin

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54. Nucleotides consist of a sugar, a phosphate group and a:

(a) Amino acid
(b) Fatty acid
(c) Nitrogenous base
(d) Glycerol

55. The sugar in DNA is:

(a) Glucose
(b) Fructose
(c) Ribose
(d) Deoxyribose

56. The sugar in RNA is:

(a) Deoxyribose
(b) Ribose
(c) Sucrose
(d) Glucose

57. Which base is found in DNA but not RNA?

(a) Thymine
(b) Guanine
(c) Uracil
(d) Adenine

58. Which base is found in RNA but not DNA?

(a) Cytosine
(b) Adenine
(c) Thymine
(d) Uracil

59. In DNA, adenine pairs with:

(a) Uracil
(b) Cytosine
(c) Thymine
(d) Guanine

60. In DNA, guanine pairs with:

(a) Uracil
(b) Adenine
(c) Cytosine
(d) Thymine

61. The two strands of DNA are:

(a) Both RNA
(b) Parallel
(c) Identical
(d) Antiparallel

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62. The double helix model of DNA was proposed by:

(a) Hooke and Brown
(b) Watson and Crick
(c) Singer and Nicolson
(d) Mendel and Morgan

63. The bond between nucleotides in a nucleic acid chain is:

(a) Hydrogen bond only
(b) Peptide bond
(c) Glycosidic bond
(d) Phosphodiester bond

64. Glycoproteins are conjugates of protein and:

(a) Phosphate only
(b) Nucleic acid
(c) Lipid
(d) Carbohydrate

65. Glycolipids are conjugates of lipid and:

(a) Carbohydrate
(b) Nucleic acid
(c) Amino acid
(d) Protein

66. Lipoproteins are important for:

(a) DNA replication
(b) Cell wall synthesis
(c) Oxygen transport only
(d) Transport of lipids in blood

67. Nucleoproteins are complexes of nucleic acid and:

(a) Carbohydrate
(b) Water
(c) Lipid
(d) Protein

68. Which of the following is a reducing sugar?

(a) Glucose
(b) Sucrose
(c) Starch
(d) Cellulose

69. The functional group –COOH is called:

(a) Amino group
(b) Phosphate group
(c) Carboxyl group
(d) Hydroxyl group

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70. The functional group –NH2 is called:

(a) Amino group
(b) Methyl group
(c) Carboxyl group
(d) Hydroxyl group

71. Which bond provides stability to complex carbohydrates?

(a) Glycosidic linkage
(b) Ionic bond only
(c) Phosphodiester bond
(d) Peptide bond

72. The C-H bond in organic molecules is important because it:

(a) Stores chemical energy
(b) Is ionic
(c) Carries genetic information only
(d) Forms the cell wall

73. Which of the following is a polysaccharide?

(a) Fructose
(b) Glucose
(c) Sucrose
(d) Cellulose

74. Which of the following is a disaccharide?

(a) Glucose
(b) Galactose
(c) Lactose
(d) Ribose

75. The loss of water during polymer formation is called:

(a) Dehydration synthesis
(b) Oxidation
(c) Hydration
(d) Hydrolysis

76. Which property of water allows insects to walk on its surface?

(a) Low specific heat
(b) Surface tension
(c) Acidity
(d) Transparency

77. The approximate specific heat capacity of water is:

(a) 1 J/g°C
(b) 4.184 J/g°C
(c) 0.5 J/g°C
(d) 10 J/g°C

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78. Which of the following is NOT a function of proteins?

(a) Catalysis
(b) Transport
(c) Immune defence
(d) Long-term energy storage as the primary role

79. Which lipid is a major component of plasma membranes?

(a) Steroid only
(b) Wax only
(c) Triacylglycerol only
(d) Phospholipid

80. The genetic information in DNA is stored in the:

(a) Histones only
(b) Sequence of bases
(c) Sugar-phosphate backbone
(d) Hydrogen bonds only

81. RNA is usually:

(a) Circular only
(b) Double-stranded
(c) Protein-coated only
(d) Single-stranded

82. Which of the following is a conjugated molecule found in cell membranes?

(a) Glycolipid
(b) Nucleotide only
(c) Glucose
(d) Amino acid

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ANSWER KEY WITH BRIEF REASONS

1. (a) Water — Water makes up 65–89% of most organisms.

2. (d) 93% — Carbohydrates, proteins, lipids and nucleic acids form about 93% of dry mass.

3. (b) Carbon — Carbon forms the backbone of organic molecules.

4. (c) Monomers — Monomers join to form polymers.

5. (d) Amino acids — Amino acids form polypeptides.

6. (a) Nucleotides — Nucleotides form DNA and RNA.

7. (c) Sharing of electrons — Covalent bonds involve shared electron pairs.

8. (c) Electrostatic attraction between oppositely charged ions — Electron transfer produces cations and anions.

9. (c) Proteins and nucleic acids — H-bonds stabilize macromolecules.

10. (a) Protein folding and lipid bilayer formation — Nonpolar groups cluster away from water.

11. (b) Removing water — Dehydration synthesis removes H and OH.

12. (a) Adding water — Water is used to break bonds.

13. (c) Condensation — A glycosidic bond forms with water removal.

14. (b) Polar and forms hydrogen bonds — Polarity allows dissolution of ions/polar molecules.

15. (c) Dissolve readily in water — Charged/polar molecules interact with water.

16. (c) Do not dissolve in water — Nonpolar molecules are excluded from water.

17. (c) Stabilizing temperature — Water resists temperature change.

18. (c) Cooling by sweating/transpiration — Evaporation removes much heat.

19. (d) Hydrogen bonding — Water molecules stick together by H-bonds.

20. (d) Cohesion and transpiration pull — Cohesion transmits the transpiration pull.

21. (d) Simple sugars — Glucose/fructose are monosaccharides.

22. (a) Isomers — They have the same formula but different structures.

42. (a) Structural and insoluble — Fibrous proteins provide support.

43. (c) Functional and soluble — Enzymes and antibodies are globular.

44. (d) Proteins — Most enzymes are globular proteins.

45. (d) Nonpolar — Hydrophobic hydrocarbon chains dominate.

46. (b) Glycerol and three fatty acids — Fats/oils are triacylglycerols.

47. (a) Solid at room temperature — No double bonds allow tight packing.

48. (c) Liquid at room temperature — Double bonds prevent tight packing.

49. (b) Cell membranes — Phospholipids form bilayers.

50. (a) Four fused carbon rings — Steroids have a four-ring skeleton.

51. (b) Steroid — Cholesterol is a steroid lipid.

52. (b) Inflammation and local signalling — Prostaglandins mediate inflammation/pain.

53. (c) Prostaglandins — Aspirin blocks prostaglandin-producing enzymes.

54. (c) Nitrogenous base — Nucleotides are the units of nucleic acids.

55. (d) Deoxyribose — DNA contains deoxyribose.

56. (b) Ribose — RNA contains ribose.

57. (a) Thymine — RNA uses uracil instead of thymine.

58. (d) Uracil — Uracil replaces thymine in RNA.

59. (c) Thymine — A-T pairing occurs in DNA.

60. (c) Cytosine — G-C pairing occurs in DNA.

61. (d) Antiparallel — DNA strands run in opposite directions.

62. (b) Watson and Crick — Watson and Crick proposed the model in 1953.

63. (d) Phosphodiester bond — Phosphodiester bonds form the backbone.

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23. (c) Glucose + glucose — Maltose is two glucose units.
24. (a) Glucose + galactose — Milk sugar contains glucose and galactose.
25. (a) Glucose + fructose — Table sugar is glucose-fructose.
26. (b) Starch — Starch stores energy in plants.

27. (b) Glycogen — Glycogen is animal storage polysaccharide.
28. (a) Plant cell walls — Cellulose strengthens plant walls.
29. (a) Fungal cell walls and arthropod exoskeletons — Chitin is a nitrogen-containing polysaccharide.
30. (c) An unbranched glucose polymer — Amylose is linear starch.
31. (a) A branched glucose polymer — Amylopectin is branched starch.
32. (a) Starch — Glycogen has more branch points than amylopectin.
33. (c) Glycosidic linkage — Sugars join by glycosidic bonds.
34. (a) Amino acid — Amino acids build proteins.

35. (a) R group — The side chain determines amino acid properties.
36. (b) Peptide bond — Peptide bonds link amino acids.
37. (d) Amino acid sequence — Primary structure is the linear sequence.

38. (c) Secondary structure — Secondary structure is local folding.
39. (a) Tertiary structure — Tertiary structure is full folding.
40. (b) Contains multiple polypeptide chains — Quaternary structure involves subunit association.
41. (a) Amino acid sequence — A single amino acid substitution alters haemoglobin.

64. (d) Carbohydrate — Glycoproteins contain sugar chains.
65. (a) Carbohydrate — Glycolipids contain sugar groups.
66. (d) Transport of lipids in blood — Lipoproteins carry hydrophobic lipids in blood.
67. (d) Protein — Chromosomes and ribosomes contain nucleoproteins.
68. (a) Glucose — Free glucose can act as a reducing sugar.
69. (c) Carboxyl group — Carboxyl groups occur in amino acids/fatty acids.
70. (a) Amino group — Amino groups occur in amino acids.

71. (a) Glycosidic linkage — Carbon-oxygen glycosidic bonds link sugars.
72. (a) Stores chemical energy — Oxidation of C-H bonds releases energy.
73. (d) Cellulose — Cellulose is a polymer of glucose.
74. (c) Lactose — Lactose is glucose + galactose.
75. (a) Dehydration synthesis — Condensation removes water.
76. (b) Surface tension — Hydrogen bonding creates high surface tension.
77. (b) 4.184 J/g°C — Water has a high specific heat.
78. (d) Long-term energy storage as the primary role — Lipids/carbohydrates are primary energy stores.
79. (d) Phospholipid — Phospholipids form the bilayer.
80. (b) Sequence of bases — Base sequence encodes information.
81. (d) Single-stranded — Most RNA is single-stranded.

82. (a) Glycolipid — Glycolipids are membrane components.

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